

Joint inference for ordered ATAC and RNA factor models

From conditional EBNM updates to a coherent joint posterior

Technical derivation for the cEBMF project

14 September 2026

Abstract

We derive a paired Gaussian matrix factorization with ordered loading priors $A_{ik} \mid A_{i,<k}$ and $R_{ij} \mid R_{i,<j}, A_i$, and HMM priors on the feature factors. The generative direction ATAC to RNA permits posterior information to flow in both directions. The loading update is a normal-means likelihood times its own conditional prior times contributions from its children. We give a prior-agnostic update and a Metropolis correction that can use any evaluable, sampleable conditional EBNM posterior. For all scalar learned registry families we also derive a common augmented Gaussian-mixture update. For the package’s CGB family, these contributions tilt an ordinary spike-and-Gaussian posterior; scalar full conditionals require one-dimensional integration, while mean-field updates additionally require expectations over uncertain parents. We give the exact conditional, a joint variational objective, normalized CGB updates and moments, HMM updates, parameter updates, and simpler binary and linear-logistic cases. A comparison with existing multi-omics methods distinguishes conditional-prior modeling from shared-subspace factorization and hierarchical neural decoders. This is a mathematical specification, not an implementation or an empirical claim that joint fitting improves both modalities.

Main conclusion. The general problem is nonconjugate, but the difficulty is localized. HMM factor smoothing remains available. The new machinery is chiefly in the loading posteriors, expected conditional-prior terms, and common objective. The current two-fitter covariate-exchange procedure does not supply these pieces automatically.

Reading guide. Sections 2–3 define the model and exact inference target. Sections 4–6 derive implementable coordinate equations. Section 5 is the general result for the learned registry, including an exact-target Gibbs option. Sections 7–8 complete the remaining updates. Sections 9–11 explain computational choices, qualifications, and usefulness. Section 12 compares the model and inference construction with related multi-omics methods and specifies the limits of capacity claims.

Contents

1	Notation and assumptions	4
2	The models being compared	4
2.1	Observation models	4
2.2	Independent and fixed-covariate fits	4
2.3	Ordered joint loading prior	5
2.4	CGB loading prior and related families	5
2.5	HMM factor priors	5
2.6	Complete joint density and inferential targets	6
3	Exact full conditionals	6
3.1	Gaussian contribution for one loading	6
3.2	Own prior and child contributions	7
4	A joint variational objective and coordinate derivation	7
4.1	Structured mean-field family	7
4.2	Derivation, not just the coordinate identity	8
4.3	Normal-means statistics with posterior uncertainty	8
5	A general update for the learned registry	9
5.1	One normalized equation for any evaluable conditional prior	9
5.2	Reusing an EBNM posterior as a reference distribution	9
5.3	An exact-target Metropolis-within-Gibbs update	10
5.4	Which probability models the registry supplies	10
5.5	A common augmented-mixture variational update	11
5.6	Sequence priors and a general factor-graph statement	12
5.7	The reusable interface implied by the mathematics	13
6	Normalized CGB updates	13
6.1	Simplifying the downstream term	13
6.2	Spike and tilted Gaussian slab	14
6.3	Positive and other slabs	14
6.4	Why the generic update is difficult	15
7	Factor HMM updates remain tractable	15
7.1	Normal-means reduction for an entire factor	15
7.2	Emissions, forward/backward and moments	16
7.3	What can be reused	16

8	Learning parameters and evaluating one objective	16
8.1	Neural gates	16
8.2	Global Gaussian slab mean and variance	17
8.3	HMM parameters on fixed component supports	17
8.4	Noise and likelihood terms	17
9	Simpler cases and a stronger variational alternative	18
9.1	Strict binary loadings: a closed-form scalar probability	18
9.2	Linear-logistic CGB: a log-concave slab	18
9.3	Keep all loadings of a cell together	18
10	Qualifications needed for a coherent implementation	19
10.1	Covariate standardization	19
10.2	The current sparsity penalty is not automatically a parameter prior	19
10.3	Ordering, rescaling, support changes and pruning	20
10.4	A mathematically defined iteration	20
11	How useful this could be	21
11.1	What the construction could provide	21
11.2	Why the supplied simulation is informative	21
11.3	The most informative first comparison	21
12	Comparison with other multi-omics methods	22
12.1	Scope and notation	22
12.2	Structural comparison	22
12.3	Observed side information and uncertain latent parents	24
12.4	Comparison with hierarchical neural models	24
12.5	Reconstruction capacity versus conditional dependence	24
12.6	Implications for inference and empirical comparison	25
A	Deriving the likelihood coefficients	26
B	Two-node binary illustration of posterior feedback	26
C	Verification and package correspondence	26

1 Notation and assumptions

We observe paired cells in two matrices, with no requirement that their features or ranks agree:

$$Y^A \in \mathbb{R}^{N \times P_A}, \quad Y^R \in \mathbb{R}^{N \times P_R}, \quad A \in \mathbb{R}^{N \times K_A}, \quad R \in \mathbb{R}^{N \times K_R}, \quad F^m \in \mathbb{R}^{P_m \times K_m} \quad (m \in \{A, R\}).$$

Here $A = L_{\text{ATAC}}$, $R = L_{\text{RNA}}$, and $L^A = A, L^R = R$. The package stores factors in the orientation $L^m(F^m)^\top$. Indices i, t, k, j denote cells, features, ATAC components and RNA components, respectively. Ω_m is the set of observed entries. Rows are aligned between modalities; feature positions need not be aligned between modalities.

- (i) The observations have independent Gaussian errors conditional on latent variables and positive precisions τ_{it}^m . This matches the notebook’s Gaussian simulation. It does not assert a count likelihood for raw RNA or ATAC.
- (ii) Loading priors act within each cell, with parameters shared across cells. External covariates, if present, are fixed and are suppressed in most formulas.
- (iii) Any centering/scaling transformation supplied to a prior network is *fixed during the inference target*. Dynamic standardization is discussed in Section 10.1; it would otherwise invalidate the cell-local Markov blankets used below.
- (iv) Factor ordering, ranks, HMM state supports and component supports are fixed unless an explicitly evaluated model-selection step changes them.
- (v) The main derivation uses the unpenalized model, equivalent to a sparsity penalty of one. Optional regularization must be defined in the common objective (Section 10.2).

Write $\varphi(x; \mu, v)$ for a Gaussian density with *variance* $v > 0$. All distributions and log densities involving a spike are interpreted on their mixed discrete/continuous space, not by assigning an infinite ordinary density to zero. Empty products equal one; empty sums equal zero.

2 The models being compared

2.1 Observation models

For $(i, t) \in \Omega_m$,

$$Y_{it}^m \mid L^m, F^m, \tau^m \sim \mathcal{N} \left(\sum_{k=1}^{K_m} L_{ik}^m F_{tk}^m, (\tau_{it}^m)^{-1} \right). \quad (2.1)$$

Denote the product likelihood by $p_m(Y^m \mid L^m, F^m, \tau^m)$. Unobserved entries contribute no likelihood term. Precisions may be known or shared within identifiable groups, such as a modality, a row or a column.

2.2 Independent and fixed-covariate fits

An independent baseline uses

$$p_0(A, R) = \prod_{i,k} g_k^A(A_{ik}) \prod_{i,j} g_j^R(R_{ij}). \quad (2.2)$$

A conditional RNA fit supplied with an estimated ATAC matrix $\hat{A}$ instead uses, for example,

$$p(R \mid \hat{A}) = \prod_i \prod_j g_j^R(R_{ij} \mid R_{i,<j}, \hat{A}_i). \quad (2.3)$$

This is a valid conditional model if its factors are normalized. Treating $\hat{A}$ as fixed does not infer ATAC loadings using RNA evidence. Even within (2.3), full RNA inference includes the effects of R_{ij} on later RNA priors. Plugging estimated earlier RNA loadings into successive EBNM calls does not include those effects.

2.3 Ordered joint loading prior

The proposed joint prior is

$$p(A, R \mid \theta) = \prod_{i=1}^N \left\{ \prod_{k=1}^{K_A} g_k^A(A_{ik} \mid A_{i,<k}; \theta_k^A) \prod_{j=1}^{K_R} g_j^R(R_{ij} \mid R_{i,<j}, A_i; \theta_j^R) \right\}. \quad (2.4)$$

It is normalized by integrating in reverse generative order. The within-cell ordering consists of all ATAC components followed by all RNA components. This is an autoregressive prior, not a causal identification result. Omitting unnecessary parent inputs gives a sparser directed graph.

For a generic loading node V_u , let $P_u = \text{pa}(V_u)$ and $C_u = \text{ch}(V_u)$. With all proposed dependencies present,

$$C_{A_{ik}} = \{A_{ih} : h > k\} \cup \{R_{ij} : 1 \leq j \leq K_R\}, \quad (2.5)$$

$$C_{R_{ij}} = \{R_{ih} : h > j\}. \quad (2.6)$$

Only actual children contribute when a network structurally omits an input.

2.4 CGB loading prior and related families

The mathematical CGB family matching `cov_gb_prior.py` is

$$\pi_u(P_u) = \text{sigmoid}\{h_{\theta_u}(S_u(P_u))\}, \quad (2.7)$$

$$Z_u \mid P_u \sim \text{Bernoulli}(\pi_u(P_u)), \quad (2.8)$$

$$V_u \mid Z_u = 0, P_u = 0, \quad V_u \mid Z_u = 1, P_u \sim \mathcal{N}(\mu_u, d_u^2), \quad (2.9)$$

where S_u denotes a fixed affine preprocessing transformation and $d_u^2 > 0$. Parameters are shared over cells for each component. In the source, `pi_2` is our slab probability π_u , `pi_1` is $1 - \pi_u$, `mu_2` is μ_u , and `sigma2_sq` is d_u^2 . The slab is signed; CGB is neither an exact Bernoulli prior nor a positive truncated-normal prior. Only its mixing probability depends on parents.

More generally, replace the slab by a normalized distribution $G_u(dv \mid P_u)$:

$$g_u(dv \mid P_u) = (1 - \pi_u(P_u))\delta_0(dv) + \pi_u(P_u)G_u(dv \mid P_u). \quad (2.10)$$

Examples include a positive truncated Gaussian, a point-exponential model, or the strict binary slab $G_u = \delta_1$. The package's intended `gbinary` family uses a positive truncated Gaussian with $d_u = \omega|\mu_u|$; this is a different family from CGB. For such a slab, the normalizing constant is $\Phi(\mu_u/d_u)$, which reduces to $\Phi(1/\omega)$ only when $\mu_u > 0$. These are model equations, not a claim that every numerical boundary case of that separate solver is audited here.

2.5 HMM factor priors

For each modality and factor, independently across (m, k) , introduce a state chain $Q_{1:P_m, k}^m$ and mixture labels $C_{1:P_m, k}^m$. Suppress (m, k) :

$$Q_1 \sim \text{Categorical}(\alpha), \quad Q_t \mid Q_{t-1} = a \sim \text{Categorical}(T_{a,\cdot}), \quad (2.11)$$

$$C_t \mid Q_t = b \sim \text{Categorical}(\rho_{b,\cdot}), \quad F_t \mid Q_t = b, C_t = c \sim G_{bc}. \quad (2.12)$$

Thus

$$p_H(F, Q, C \mid \psi) = \alpha_{Q_1} \prod_{t=2}^{P_m} T_{Q_{t-1}, Q_t} \prod_{t=1}^{P_m} \rho_{Q_t, C_t} G_{Q_t, C_t}(dF_t). \quad (2.13)$$

Rows of T and ρ , and α , sum to one. The signed HMM uses $G_{bc} = \mathcal{N}(c_b, v_c)$ for $v_c > 0$, and $G_{bc} = \delta_{c_b}$ for $v_c = 0$. The positive model uses

$$G_{bc}^+(df) = \frac{\varphi(f; c_b, v_c) \mathbf{1}\{f > 0\}}{\Phi(c_b/\sqrt{v_c})} df \quad (v_c > 0), \quad (2.14)$$

with nonnegative atoms when $v_c = 0$. The negative model is the reflection of the positive model. A designated null state can be exactly δ_0 . This captures the probability model of the current HMM implementation; its automatic grids, moving centers and pruning are estimation procedures.

Each chain follows the existing feature order with unit spacing. There is no cross-modality feature alignment, distance-dependent transition, or learned dependence on $\mathbf{X}_f$ in this HMM specification. Unrelated sequences require separate chains or an explicitly specified reset.

2.6 Complete joint density and inferential targets

Let $\mathcal{H}$ collect all factor chains and mixture labels, and let $\Theta = (\theta, \psi, \tau)$ collect parameters. Then

$$\begin{aligned} p(Y^A, Y^R, A, R, F^A, F^R, \mathcal{H} \mid \Theta) &= p_A(Y^A \mid A, F^A, \tau^A) p_R(Y^R \mid R, F^R, \tau^R) \\ &\times p(A, R \mid \theta) \prod_{m \in \{A, R\}} \prod_{k=1}^{K_m} p_H(F_k^m, \mathcal{H}_k^m \mid \psi_k^m). \end{aligned} \quad (2.15)$$

An empirical Bayes analysis estimates Θ and targets the joint latent posterior conditional on that estimate. A fully Bayesian analysis additionally specifies $p(\Theta)$ and infers parameter uncertainty. Joint latent inference does not require fully Bayesian treatment of every hyperparameter. Conversely, optimizing all latent variables jointly as point estimates is joint MAP, not posterior uncertainty propagation. With spikes, MAP also needs a precise discrete/continuous parameterization; comparing an atom's mass to a continuous density is not an invariant decision rule.

3 Exact full conditionals

3.1 Gaussian contribution for one loading

Hold other variables and parameters fixed and set

$$e_{it, -k}^m = Y_{it}^m - \sum_{h \neq k} L_{ih}^m F_{th}^m, \quad (3.1)$$

$$a_{ik}^m = \sum_{t: (i, t) \in \Omega_m} \tau_{it}^m (F_{tk}^m)^2, \quad b_{ik}^m = \sum_{t: (i, t) \in \Omega_m} \tau_{it}^m F_{tk}^m e_{it, -k}^m. \quad (3.2)$$

Terms depending on a candidate $v = L_{ik}^m$ reduce exactly to

$$\ell_u(v) = -\frac{1}{2} a_u v^2 + b_u v. \quad (3.3)$$

If $a_u > 0$, define $s_u^2 = 1/a_u$, $x_u = b_u/a_u$. Then $\exp \ell_u(v) \propto \varphi(x_u; v, s_u^2)$. If $a_u = 0$, there is no direct data information and $b_u = 0$; retain the constant likelihood rather than inventing information through an epsilon. Children can still inform the loading in this case.

3.2 Own prior and child contributions

Use $P_c[v]$ for a child's parent vector with the target replaced by v . The exact conditional is

$$p(dv \mid \text{rest}, Y, \Theta) \propto \exp\{\ell_u(v)\} g_u(dv \mid P_u) \prod_{c \in \mathcal{C}_u} g_c(V_c \mid P_c[v]). \quad (3.4)$$

For continuous slabs a zero child uses its spike probability, while a nonzero child uses its slab weight times its slab density. This interpretation also follows by introducing the indicators in (2.9). In particular,

$$p(dA_{ik} \mid \text{rest}, Y, \Theta) \propto e^{\ell_{ik}^A(A_{ik})} g_k^A(dA_{ik} \mid A_{i,<k}) \times \prod_{h>k} g_h^A(A_{ih} \mid A_{i,<h}) \prod_j g_j^R(R_{ij} \mid R_{i,<j}, A_i), \quad (3.5)$$

$$p(dR_{ij} \mid \text{rest}, Y, \Theta) \propto e^{\ell_{ij}^R(R_{ij})} g_j^R(dR_{ij} \mid R_{i,<j}, A_i) \prod_{h>j} g_h^R(R_{ih} \mid R_{i,<h}, A_i). \quad (3.6)$$

Only terms depending on the target need evaluation. The RNA likelihood is constant in (3.5) when R, F^R are fixed; its evidence reaches ATAC through the inferred RNA loadings and the RNA conditional priors. This is posterior feedback under the original ATAC-to-RNA joint prior.

Sampling these full conditionals, together with valid updates for all remaining unknowns, gives a Gibbs or Metropolis-within-Gibbs scheme for one joint target. Replacing samples by conditional means is not Gibbs sampling. Fitting two unrelated reverse-direction conditional priors does not ensure that they are full conditionals of (2.15). Re-estimating parameters during sampling also changes the target; a Bayesian sampler must update those parameters under their conditional distributions, or an EB analysis must separate estimation from inference conditional on the fitted parameters.

4 A joint variational objective and coordinate derivation

4.1 Structured mean-field family

Keep each factor's HMM sequence intact, but initially factorize loading nodes:

$$q = \left\{ \prod_{i,k} q_{ik}^A(A_{ik}) \prod_{i,j} q_{ij}^R(R_{ij}) \right\} \prod_{m,k} q_{F,k}^m(F_k^m, \mathcal{H}_k^m). \quad (4.1)$$

Each loading factor can contain an exact spike and an unrestricted continuous slab. The approximation loses posterior dependence between different loading nodes, but retains their uncertainty and includes every joint-prior term in the objective. The factor sequences are independent across (m, k) under q , not independent across feature positions within a sequence.

For fixed Θ , the evidence lower bound is

$$\mathcal{L}(q, \Theta) = \mathbb{E}_q \log p(Y, A, R, F, \mathcal{H} \mid \Theta) - \mathbb{E}_q \log q. \quad (4.2)$$

It equals $\log p(Y \mid \Theta) - \text{KL}(q \parallel p(\cdot \mid Y, \Theta))$. For this model it can be organized as

$$\begin{aligned} \mathcal{L} = & \sum_m \mathbb{E}_q \log p_m(Y^m \mid L^m, F^m, \tau^m) + \sum_u \mathbb{E}_q \log g_u(V_u \mid P_u) \\ & + \sum_u H(q_u) - \sum_{m,k} \text{KL}(q_{F,k}^m \parallel p_H^m(\cdot \mid \psi_k^m)). \end{aligned} \quad (4.3)$$

Every conditional prior appears *once*. It may enter the updates of several variables because it depends on each of them. This is not double counting in the objective. If parameters have a prior and are point-estimated, add $\log p(\Theta)$ for a MAP parameter objective. A variational distribution over parameters requires its own expected prior and entropy terms instead.

4.2 Derivation, not just the coordinate identity

Holding q_{-u} fixed, collect $f_u(v) = \mathbb{E}_{q_{-u}} \log p(Y, V_u = v, V_{-u}, F, \mathcal{H} \mid \Theta)$. The terms varying with q_u are

$$\int q_u(v) f_u(v) d\nu_u(v) - \int q_u(v) \log q_u(v) d\nu_u(v) = -\text{KL}(q_u \| q_u^*) + \log Z_u, \quad (4.4)$$

where $q_u^*(v) = \exp f_u(v) / Z_u$, with a suitable mixed reference measure ν_u . Thus the unrestricted coordinate optimum is

$$\log q_u^*(v) = C + \mathbb{E}_{q_{-u}} \log p(Y, V, F, \mathcal{H} \mid \Theta). \quad (4.5)$$

This standard identity is reviewed by Blei et al. [2]; the remaining expressions below follow by expanding this particular joint model. Exact coordinate maximization cannot decrease (4.2). A Gaussian approximation to a tilted slab, noisy quadrature or incomplete network optimization does not inherit that guarantee automatically.

4.3 Normal-means statistics with posterior uncertainty

Define $m_{ik}^L = \mathbb{E} L_{ik}$, $M_{ik}^L = \mathbb{E} L_{ik}^2$, $m_{tk}^F = \mathbb{E} F_{tk}$, $M_{tk}^F = \mathbb{E} F_{tk}^2$, separately by modality. Using (4.1) and fixed precisions gives

$$a_{ik}^m = \sum_{t:(i,t) \in \Omega_m} \tau_{it}^m M_{tk}^{F,m}, \quad (4.6)$$

$$b_{ik}^m = \sum_{t:(i,t) \in \Omega_m} \tau_{it}^m m_{tk}^{F,m} \left(Y_{it}^m - \sum_{h \neq k} m_{ih}^{L,m} m_{th}^{F,m} \right). \quad (4.7)$$

The coefficient of v^2 uses a *second moment*, not the squared mean. For an independent variational precision factor replace τ by $\mathbb{E}\tau$. The remainder of the coordinate update is

$$\boxed{\log q_u^*(v) = C - \frac{1}{2} a_u v^2 + b_u v + \mathbb{E} \log g_u(v \mid P_u) + D_u(v), \quad D_u(v) = \sum_{c \in \mathcal{C}_u} \mathbb{E} \log g_c(V_c \mid P_c[v])}. \quad (4.8)$$

Expectations integrate all other uncertain arguments, never the candidate v . Evaluating a network at posterior means generally differs from the expected log probability in this equation.

The current conditional EBNM approximation effectively uses

$$q_u^{\text{plug}}(dv) \propto \varphi(x_u; v, s_u^2) g_u(dv \mid \mathbb{E} P_u). \quad (4.9)$$

It both omits D_u and replaces uncertain predictors by their means. An ordinary local EBNM marginal likelihood cannot be substituted for all the joint-prior and entropy terms in (4.3) after this change.

5 A general update for the learned registry

5.1 One normalized equation for any evaluable conditional prior

Assume that the conditional priors admit densities with respect to specified reference measures with fixed supports, and that the following normalizer is positive and finite. These assumptions cover the scalar learned priors in the registry with fixed component supports and positive variances. Define

$$U_u(v) = -\frac{1}{2}a_u v^2 + b_u v + \mathbb{E} \log g_u(v \mid P_u) + \sum_{c \in \mathcal{C}_u} \mathbb{E} \log g_c(V_c \mid P_c[v]). \quad (5.1)$$

Then the general mean-field coordinate update and its moments are

$$q_u^*(dv) = \frac{e^{U_u(v)} \nu_u(dv)}{\int e^{U_u(w)} \nu_u(dw)}, \quad \mathbb{E}_{q_u^*} f(V_u) = \frac{\int f(v) e^{U_u(v)} \nu_u(dv)}{\int e^{U_u(v)} \nu_u(dv)}. \quad (5.2)$$

For a spike at zero plus a continuous density, $\nu = \delta_0 + \text{Lebesgue}$. The density at the atom is its probability mass. Its value on the continuous part is the continuous mixture density; one must not add a Gaussian density evaluated at zero to the atom's mass. Other finite atoms are handled by adding their separate mass terms to the integrals. Priors with moving supports or singular parameter-dependent transformations need separate treatment.

Equation (5.2) is a *general update*, but not a universal closed-form solver. It requires conditional log-prior evaluation for arbitrary candidate values and uncertain parent inputs. A function returning only `post_mean`, `post_mean2` and a local marginal likelihood does not expose enough information to evaluate it.

5.2 Reusing an EBNM posterior as a reference distribution

Freeze prior parameters and choose a reference parent value $p_\star$. Let an ordinary normal-means posterior be

$$q_{0u}(dv) \propto \varphi(x_u; v, s_u^2) g_u(dv \mid p_\star). \quad (5.3)$$

Suppose it is positive everywhere the desired update has positive mass or density. With all densities defined against the same reference measure, put

$$C_u(v) = \mathbb{E} \log g_u(v \mid P_u) - \log g_u(v \mid p_\star) + D_u(v). \quad (5.4)$$

Then

$$q_u^*(dv) = \frac{q_{0u}(dv) e^{C_u(v)}}{\mathbb{E}_{q_{0u}} e^{C_u(V_u)}}. \quad (5.5)$$

Thus quadrature or importance sampling from an existing conditional posterior can in principle support every scalar learned family, including EMDN with parent-dependent means and variances. For samples $v_b \sim q_{0u}$, a self-normalized approximation is $\mathbb{E} f(V_u) \approx \sum_b \tilde{w}_b f(v_b)$, $\tilde{w}_b \propto e^{C_u(v_b)}$. This finite-sample ratio is not unbiased, can have highly variable weights, and needs support and tail checks. If spike probability is small, integrating its mass separately is preferable to hoping it appears in a finite sample. The identity is exact; this numerical procedure need not be.

5.3 An exact-target Metropolis-within-Gibbs update

There is an especially simple general route if sampling is acceptable. For an exact scalar full conditional, all other latent variables are fixed. Compute a_u, b_u from their *realized values* as in Section 3, not from the variational moments in (4.6)–(4.7), and take $p_\star = P_u$. Then the correction in (5.5) is just the exact child log likelihood

$$D_u^G(v) = \sum_{c \in \mathcal{C}_u} \log g_c(V_c \mid P_c[v]). \quad (5.6)$$

Propose $v' \sim q_{0u}$, the *ordinary conditional normal-means posterior* under the fixed fitted prior. From the current value v , accept with probability

$$\mathcal{A}(v, v') = \min \left\{ 1, \exp [D_u^G(v') - D_u^G(v)] \right\}. \quad (5.7)$$

To verify it, write the target $t(v) \propto q_{0u}(v)e^{D_u^G(v)}$. The independence Metropolis ratio [1] is $t(v')q_{0u}(v)/[t(v)q_{0u}(v')] = e^{D_u^G(v') - D_u^G(v)}$. The cancellation works for both atom masses and continuous densities on the mixed measure. It does not require differentiation, a Gaussian slab, or a normalizing integral for the tilted target.

For a fixed-parent Gaussian mixture, sampling q_{0u} is straightforward: sample a component using weights proportional to its prior weight times its normal-means evidence, then sample its Gaussian posterior or atom. All scalar learned registry families below have that form once their parameters and parents are fixed. The existing interface would need to expose this posterior, not merely its moments. If a different proposal r is used, the acceptance ratio must include the uncanceled target/proposal terms.

One or several such invariant transitions per coordinate yields Metropolis-within-Gibbs, provided the overall chain has the required support and ergodicity. This is exact-target MCMC, not an exact independent posterior draw in finitely many steps. Strong child constraints may make acceptance poor and mixing slow. For a complete sampler, factor HMM blocks must also be *sampled* using forward-filter/backward-sampling and component/effect draws, with pseudo-data formed from realized current loadings, rather than replaced by posterior means. Parameters must be fixed for conditional EB inference or updated under their Bayesian conditionals. Refitting an EBNM prior on each proposal destroys this cancellation and fixed target unless that parameter update is separately justified.

5.4 Which probability models the registry supplies

The mapping below is based on the current `LearnedPriorType` and solver sources. A Gaussian whose mean is constrained positive still has signed support. A sharpening step during estimation is not itself a different normalized likelihood or a proof of a joint-objective ascent step.

Registry name	Conditional prior	Joint-update consequence
cash	Spike at zero plus zero-centered Gaussian scale mixture; neural mixture weights.	Fixed-component mixture formulas apply; child messages involve the full mixture density or explicit component labels.
1cash	Same scale mixture; linear softmax weights.	Same generic mixture update, with a simpler gate.
po_1cash	Same scale mixture; ordered-logistic probabilities.	Same generic mixture update; respect ordered cut-point constraints.

Registry name	Conditional prior	Joint-update consequence
cgb	Spike plus one signed Gaussian; neural slab probability, global mean and variance.	Section 6 gives its simpler scalar spike/tilted-Gaussian update.
cgb_sharp	Same probability family as CGB at fixed parameters; variance sharpening during fitting.	Conditional inference has the CGB form; the sharpening rule needs a separate objective justification.
cgb_sharp_2	Spike plus two signed Gaussians with global positive/negative mean constraints and neural weights.	General mixture form; sign-constrained means do not truncate the slabs.
emdn	Gaussian mixture whose weights, means and variances depend on covariates. No exact spike.	Include parent effects on every mixture parameter in child messages.
spiked_emdn	Exact zero spike plus EMDN Gaussian slabs.	Same EMDN message with a separate spike mass.
hmm, hmm_pos, hmm_neg	Sequence priors with latent Markov states; ignore covariates.	Treat the dependent sequence as a block. They are not scalar covariate-density priors merely because they appear in the learned registry.

For scalar priors, automatic component grids should be fixed while deriving one target, or changed only through evaluated model proposals. Conditional variances must remain positive for continuous components. Fixed-component weights can change with parents without moving the underlying atom support.

5.5 A common augmented-mixture variational update

All scalar entries in the table can be represented, at the probability-model level, using a component label B_u :

$$p(B_u = h, V_u | P_u) = w_{uh}(P_u) G_{uh}(V_u | P_u), \quad \sum_h w_{uh}(P_u) = 1. \quad (5.8)$$

The labels in this subsection are distinct from the HMM mixture labels. For continuous components, write $G_{uh} = \mathcal{N}(\mu_{uh}(P_u), d_{uh}^2(P_u))$; an optional component is the fixed zero atom. Use the variational family $\prod_u q_u(V_u, B_u) \prod_{m,k} q_{F,k}^m$, so that the value and its own label are *not* forced independent. Parents enter the model through their values, not through these auxiliary labels.

This is a different variational family on an augmented joint model. In general,

$$\mathbb{E} \log \sum_h w_h G_h \neq \log \sum_h \exp\{\mathbb{E} \log(w_h G_h)\}.$$

The augmented update must not be advertised as the exact collapsed coordinate (5.2) when parents are uncertain. Both target the same generative marginal model, but their variational approximations differ.

Let $r_{ch} = q(B_c = h)$, and for a continuous child component let $m_{ch} = \mathbb{E}(V_c | B_c = h)$, $M_{ch} = \mathbb{E}(V_c^2 | B_c = h)$. The downstream contribution in the augmented family is

$$D_u^{\text{aug}}(v) = \sum_{c \in \mathcal{C}_u} \sum_h r_{ch} \mathbb{E}_{P_c \setminus V_u} [\log w_{ch}(P_c[v]) + J_{ch}(P_c[v])], \quad (5.9)$$

where, for a Gaussian child component,

$$J_{ch}(p) = -\frac{1}{2} \log(2\pi d_{ch}^2(p)) - \frac{M_{ch} - 2\mu_{ch}(p)m_{ch} + \mu_{ch}^2(p)}{2d_{ch}^2(p)}. \quad (5.10)$$

For the fixed spike component, $J_{c0} = 0$ on its supported event. This uses the per-node mean-field independence from parents while preserving $q(V_c, B_c)$. It handles parent-dependent EMDN means and variances, not only weight-dependent CGB gates.

For each continuous component of the target, define

$$A_h = a_u + \mathbb{E}[d_{uh}^{-2}(P_u)], \quad (5.11)$$

$$B_h = b_u + \mathbb{E}[\mu_{uh}(P_u)d_{uh}^{-2}(P_u)], \quad (5.12)$$

$$C_h = \mathbb{E} \log w_{uh}(P_u) - \frac{1}{2} \mathbb{E} \log(2\pi d_{uh}^2(P_u)) - \frac{1}{2} \mathbb{E}[\mu_{uh}^2(P_u)d_{uh}^{-2}(P_u)]. \quad (5.13)$$

Assume these expectations are finite and $A_h > 0$. Put $v_h = 1/A_h$, $m_h = B_h/A_h$, $I_{rh} = \int v^r \varphi(v; m_h, v_h) e^{D_u^{\text{aug}}(v)} dv$. Completing the square gives unnormalized component masses

$$W_h = \exp \left\{ C_h + \frac{B_h^2}{2A_h} \right\} \sqrt{\frac{2\pi}{A_h}} I_{0h}, \quad h \text{ continuous}, \quad (5.14)$$

$$W_0 = \exp\{\mathbb{E} \log w_{u0}(P_u) + D_u^{\text{aug}}(0)\} \quad \text{if a zero atom is present.} \quad (5.15)$$

There is no normal-means evidence constant in (5.15) because the common likelihood factor $\exp \ell_u(v)$, rather than a normalized Gaussian in x , was used for *all* components. After setting $r_{uh} = W_h / \sum_b W_b$,

$$q_u^*(dv, B_u = h) = r_{uh} \frac{\varphi(v; m_h, v_h) e^{D_u^{\text{aug}}(v)}}{I_{0h}} dv \quad (h \text{ continuous}), \quad (5.16)$$

with mass r_{u0} at the spike. Therefore $\mathbb{E}V_u = \sum_{h \text{ cont}} r_{uh} I_{1h} / I_{0h}$ and $\mathbb{E}V_u^2 = \sum_{h \text{ cont}} r_{uh} I_{2h} / I_{0h}$. This is a common tilted-Gaussian-mixture update for every scalar learned registry family, including EMDN. It remains nonconjugate through the downstream tilt and uncertain-parent expectations, but no new family-specific derivation is needed for each choice of gate or covariate-dependent Gaussian parameters.

For fixed Gaussian components, simplify the expectations in A_h, B_h, C_h to their constant values and expected log weights. For CGB, the zero and Gaussian components have disjoint supports up to a measure-zero event, so these equations reduce to Section 6. For overlapping Gaussian mixtures, component augmentation genuinely changes the chosen mean-field approximation.

The augmented parameter update is likewise general:

$$\theta_u^{\text{new}} \in \arg \max_{\theta_u} \sum_i \mathbb{E}_{q(P_{iu})q(V_{iu}, B_{iu})} [\log w_{uB_{iu}}(P_{iu}; \theta_u) + \log G_{uB_{iu}}(V_{iu} | P_{iu}; \theta_u)] + \log p(\theta_u). \quad (5.17)$$

For EMDN this trains weights, means and variances together using latent posterior expectations; for fixed-component mixtures only the weight-network terms vary. It is not generally the old isolated EBNM marginal-likelihood training call. The augmented ELBO must retain component-label entropy and expected complete-data prior terms consistently.

5.6 Sequence priors and a general factor-graph statement

For a block V_B , the same principle is

$$\log q_B^*(v_B) = C + \mathbb{E}_{q_{-B}} \sum_{f: \text{scope}(f) \cap B \neq \emptyset} \log f(v). \quad (5.18)$$

Sum every joint factor touching the block, not just scalar parent/child terms. This covers HMM priors without falsely making their coordinates independent. In the requested model the HMM

is on F , where it has no loading children, so Section 7 gives the direct block update. If an HMM were placed on L , it would couple ordered cells; those states and neighbors must be included. When its downstream correction factorizes over positions, smoothing can still use emissions obtained by one-dimensional component integrals with the extra tilt. More general cross-position corrections need a different block solver. An HMM posterior proposal can also be Metropolis-corrected using the full block downstream log-likelihood difference, provided the proposal can be sampled and the required support condition holds.

5.7 The reusable interface implied by the mathematics

A future general solver would need, conceptually:

1. Frozen prior parameters, normalized conditional log densities and explicit atom masses at arbitrary parent values, with fixed preprocessing.
2. A representation of the prior and posterior components, and either posterior sampling or numerical integration support.
3. Expected log-prior and parameter-gradient evaluations under uncertain parents and child values, if using variational inference.
4. A joint dependency graph and one consistent objective, including any component-label entropy, HMM block terms and declared penalties.

These are mathematical requirements, not proposed code changes in this task. The current registry can select the component family; its moment-returning `fit` contract alone is insufficient for a universal joint update. The same scalar formulas apply to a feature-factor entry if a learned conditional prior is placed on F , after substituting its factor-side Gaussian coefficients and its actual parent/child graph. With fixed external feature covariates and no factor children, the update reduces to the ordinary conditional EBNM problem. Latent self-covariates on F introduce the same expected-parent and child-message requirements as on L .

6 Normalized CGB updates

6.1 Simplifying the downstream term

Let $\gamma_c = q(Z_c = 1)$ denote a child’s posterior slab probability. Because CGB’s μ_c, d_c^2 are global and independent of parents, its slab-density term is constant in the parent’s candidate v . Under (4.1), we may therefore use the following representative of D_u , dropping additive constants:

$$D_u(v) = \sum_{c \in \mathcal{C}_u} [(1 - \gamma_c) \mathbb{E} \log(1 - \pi_c(P_c[v])) + \gamma_c \mathbb{E} \log \pi_c(P_c[v])]. \quad (6.1)$$

The expectation is only over the child’s other parents. This formula uses independence between a child’s loading and its parents under the chosen variational family. It would require modification for a structured joint loading approximation. A parent-dependent slab mean or variance also adds expected child slab-density terms; they cannot then be dropped.

For an exact scalar Gibbs conditional, use actual parents, no expectations, and $\gamma_c = Z_c$ in (6.1). Conditional on the other loadings the same normalized formulas below apply. For a continuous slab, $Z_c = \mathbf{1}\{V_c \neq 0\}$ almost surely.

Define the target’s expected log gate weights

$$\eta_{u0} = \mathbb{E} \log(1 - \pi_u(P_u)), \quad \eta_{u1} = \mathbb{E} \log \pi_u(P_u). \quad (6.2)$$

Their exponentials need not sum to one. Replacing them by $1 - \pi_u(\mathbb{E}P_u)$ and $\pi_u(\mathbb{E}P_u)$ is an additional approximation.

6.2 Spike and tilted Gaussian slab

Suppress the node index and assume $a > 0$. Gaussian multiplication gives

$$v_0 = (s^{-2} + d^{-2})^{-1}, \quad m_0 = v_0(x/s^2 + \mu/d^2), \quad (6.3)$$

$$\varphi(x; v, s^2) \varphi(v; \mu, d^2) = \varphi(x; \mu, s^2 + d^2) \varphi(v; m_0, v_0). \quad (6.4)$$

Let

$$I_r = \int_{\mathbb{R}} v^r \varphi(v; m_0, v_0) \exp\{D(v)\} dv, \quad r = 0, 1, 2. \quad (6.5)$$

The unnormalized posterior masses and their normalization are

$$w_0 = e^{\eta_0} \varphi(x; 0, s^2) e^{D(0)}, \quad (6.6)$$

$$w_1 = e^{\eta_1} \varphi(x; \mu, s^2 + d^2) I_0, \quad \gamma = \frac{w_1}{w_0 + w_1}. \quad (6.7)$$

Consequently the exact optimum within the unrestricted scalar family is

$$q^*(dv) = (1 - \gamma) \delta_0(dv) + \gamma \frac{\varphi(v; m_0, v_0) e^{D(v)}}{I_0} dv. \quad (6.8)$$

Its moments are

$$\mathbb{E}V = \gamma I_1 / I_0, \quad \mathbb{E}V^2 = \gamma I_2 / I_0, \quad \text{Var}(V) = \gamma I_2 / I_0 - (\gamma I_1 / I_0)^2. \quad (6.9)$$

The posterior null probability is $1 - \gamma$. These moments alone do not specify a general non-Gaussian slab. To evaluate future expected log gates, the algorithm must retain a representation of that slab (for example, quadrature nodes and weights, or samples), or declare an additional approximation.

An equivalent log-odds form isolates the downstream correction:

$$\text{logit } \gamma = \eta_1 - \eta_0 + \log \frac{\varphi(x; \mu, s^2 + d^2)}{\varphi(x; 0, s^2)} + \log I_0 - D(0). \quad (6.10)$$

With no children, choose $D = 0$: $I_0 = 1, I_1 = m_0, I_2 = m_0^2 + v_0$, recovering the usual spike-and-Gaussian posterior, with expected-log gate weights for uncertain parents. With fixed parents this is ordinary CGB EBNM. The last RNA component has no children, but its own parent uncertainty remains.

If $a = 0$, replace the base Gaussian by $\mathcal{N}(\mu, d^2)$ directly and use $w_0 = e^{\eta_0 + D(0)}, w_1 = e^{\eta_1} I_0$. This is also the appropriate limit of the likelihood-kernel derivation. Compute weights in log space.

6.3 Positive and other slabs

For a positive Gaussian slab, the ordinary un-tilted slab posterior is $\mathcal{N}_+(m_0, v_0)$. Its marginal normal-means likelihood is

$$B_+(x) = \varphi(x; \mu, s^2 + d^2) \frac{\Phi(m_0/\sqrt{v_0})}{\Phi(\mu/d)}. \quad (6.11)$$

Use $I_r^+ = \mathbb{E}_{V \sim \mathcal{N}_+(m_0, v_0)}[V^r e^{D(V)}]$, and replace $\varphi(x; \mu, s^2 + d^2) I_0$ in (6.7) by $B_+(x) I_0^+$. The moment formula is otherwise unchanged. This is a possible positive *loading* model; selecting `hmm_pos` for F does not make the existing CGB loading slab positive.

For a parent-independent exponential slab with rate $\lambda > 0$, its un-tilted posterior is $\mathcal{N}_+(x - \lambda s^2, s^2)$ and its evidence is

$$B_{\text{exp}}(x) = \lambda \exp\{-\lambda x + \lambda^2 s^2 / 2\} \Phi(x/s - \lambda s). \quad (6.12)$$

The same spike/slab evidence-times-tilt construction applies. For general parent-dependent slabs, start from (4.8); neither this Gaussian base nor the simplified CGB message is generally available.

6.4 Why the generic update is difficult

For a scalar full conditional, $D(v)$ can be evaluated with all other variables fixed, so normalization is one-dimensional, even for a neural gate. For a variational update, evaluating $D(v)$ and η_z also requires expectations over uncertain parent vectors. There is usually no analytic formula for these logistic/neural expectations. Deterministic quadrature over all parents scales poorly; Monte Carlo replaces it with sampling error.

A possible numerical calculation draws parents from their current variational factors, reuses those draws across candidate v , averages *log* gate probabilities, and then integrates (6.5) in one dimension. It must preserve the exact spike weight and check normalization and tails. Exponentiating an unbiased estimate of $\mathbb{E} \log g$ is not an unbiased estimate of its exponential. Finite Monte Carlo coordinate steps are therefore approximations to CAVI, not exact monotone updates. Arbitrary neural gates can also produce multiple slab modes.

7 Factor HMM updates remain tractable

7.1 Normal-means reduction for an entire factor

Because loading priors do not depend on F , no loading-child term enters a factor update. Under (4.1), define

$$a_{tk}^{F,m} = \sum_{i:(i,t) \in \Omega_m} \tau_{it}^m M_{ik}^{L,m}, \quad (7.1)$$

$$b_{tk}^{F,m} = \sum_{i:(i,t) \in \Omega_m} \tau_{it}^m m_{ik}^{L,m} \left(Y_{it}^m - \sum_{h \neq k} m_{ih}^{L,m} m_{th}^{F,m} \right), \quad (7.2)$$

$$(s_{tk}^{F,m})^2 = (a_{tk}^{F,m})^{-1}, \quad x_{tk}^{F,m} = b_{tk}^{F,m} / a_{tk}^{F,m}. \quad (7.3)$$

For positive $a_{tk}^{F,m}$,

$$q_{F,k}^{m*}(dF, Q, C) \propto p_H(dF, Q, C \mid \psi_k^m) \prod_t \varphi(x_{tk}^{F,m}; F_t, (s_{tk}^{F,m})^2). \quad (7.4)$$

This is precisely an HMM normal-means smoothing problem. If a location has $a_{tk}^{F,m} = 0$, its emission likelihood is one: the chain still supplies information. A finite-SE-only solver needs an explicit missing-emission extension for that case; a large finite SE is only an approximation.

7.2 Emissions, forward/backward and moments

Suppress (m, k) and write x_t, s_t^2 . For state b , component c ,

$$e_{tbc} = \int \varphi(x_t; f, s_t^2) G_{bc}(df), \quad B_t(b) = \sum_c \rho_{bc} e_{tbc}. \quad (7.5)$$

For a signed Gaussian component, $e_{tbc} = \varphi(x_t; c_b, s_t^2 + v_c)$; for an atom, $e_{tbc} = \varphi(x_t; c_b, s_t^2)$. For a positive continuous component, put $v_* = (s_t^{-2} + v_c^{-1})^{-1}$, $m_* = v_*(x_t/s_t^2 + c_b/v_c)$, and use

$$e_{tbc}^+ = \varphi(x_t; c_b, s_t^2 + v_c) \frac{\Phi(m_*/\sqrt{v_*})}{\Phi(c_b/\sqrt{v_c})}. \quad (7.6)$$

Compute state smoothing using scaled or log-space versions of

$$f_1(b) = \alpha_b B_1(b), \quad f_t(b) = B_t(b) \sum_a f_{t-1}(a) T_{ab}, \quad (7.7)$$

$$g_{P_m}(b) = 1, \quad g_t(a) = \sum_b T_{ab} B_{t+1}(b) g_{t+1}(b). \quad (7.8)$$

After normalization,

$$\Gamma_t(b) \propto f_t(b) g_t(b), \quad (7.9)$$

$$\Xi_t(a, b) \propto f_{t-1}(a) T_{ab} B_t(b) g_t(b), \quad t \geq 2, \quad (7.10)$$

$$W_{tbc} = \Gamma_t(b) \frac{\rho_{bc} e_{tbc}}{B_t(b)}. \quad (7.11)$$

Each Γ_t , Ξ_t , and W_t sums to one over its state indices. Within a component, the conditional effect posterior is a Gaussian $\mathcal{N}(m_*, v_*)$, its positive truncation, or the corresponding atom. For the positive truncation let $z = m_*/\sqrt{v_*}$ and $r(z) = \varphi(z; 0, 1)/\Phi(z)$. Then

$$\mathbb{E}(F_t \mid b, c, x) = m_* + \sqrt{v_*} r(z), \quad \mathbb{E}(F_t^2 \mid b, c, x) = m_*^2 + v_* + m_* \sqrt{v_*} r(z). \quad (7.12)$$

Weight these moments by W_{tbc} . Stable tail evaluation is necessary when z is very negative. Reflection gives the negative model's moments. If requested, $\text{lfsr}_t = \min\{q(F_t \leq 0), q(F_t \geq 0)\}$; an atom at zero is in both tails, so in either one-sided model this equals $q(F_t = 0)$.

7.3 What can be reused

For fixed HMM parameters, the existing forward/backward and emission-moment machinery solves (7.4). Learning HMM parameters by increasing its normal-means marginal likelihood, while optimizing the corresponding factor posterior and holding the rest fixed, can serve as an EB block update. This requires an increase in the relevant common objective; it is not a guarantee for every heuristic initialization, center move or pruning proposal. State-support changes need particular care (Section 10.3).

8 Learning parameters and evaluating one objective

8.1 Neural gates

For each component, hold the variational distribution fixed and maximize

$$\sum_i [(1 - \gamma_{iu}) \mathbb{E} \log(1 - \pi_{\theta_u}(P_{iu})) + \gamma_{iu} \mathbb{E} \log \pi_{\theta_u}(P_{iu})] + \log p(\theta_u). \quad (8.1)$$

Omit the last term for unpenalized EB. This is expected Bernoulli cross-entropy with soft responses and uncertain predictors. Sampling parent values supplies a stochastic estimate of its gradient. During this parameter step, q is fixed. During a *loading* step, dependence of all relevant child probabilities on that loading must be retained. These are distinct operations. The current isolated normal-means network fit does not by itself implement this global alternating optimization.

8.2 Global Gaussian slab mean and variance

Let $m_{iu} = \mathbb{E}V_{iu}$, $M_{iu} = \mathbb{E}V_{iu}^2$, and $G_u = \sum_i \gamma_{iu} > 0$. Without parameter priors, the M-step is

$$\boxed{\mu_u^{\text{new}} = \frac{\sum_i m_{iu}}{G_u}, \quad d_u^2{}^{\text{new}} = \frac{\sum_i \{M_{iu} - 2\mu_u^{\text{new}}m_{iu} + (\mu_u^{\text{new}})^2\gamma_{iu}\}}{G_u}}. \quad (8.2)$$

These are posterior latent moments; there is no subtraction of the normal-means observation variance here. Formula (8.2) is for a free Gaussian slab, not for a truncated slab or the constraint $d = \omega|\mu|$. Those cases require their own expected log-density optimization, including normalizing constants. If $G_u = 0$, slab parameters are unidentified and an explicit convention or prior is needed. A variance floor changes the domain of the optimization and should be stated.

8.3 HMM parameters on fixed component supports

For one chain, standard expected-count maximizers are

$$\alpha_b^{\text{new}} = \Gamma_1(b), \quad T_{ab}^{\text{new}} = \frac{\sum_{t=2}^{P_m} \Xi_t(a, b)}{\sum_{t=2}^{P_m} \Gamma_{t-1}(a)}, \quad \rho_{bc}^{\text{new}} = \frac{\sum_t W_{tbc}}{\sum_t \Gamma_t(b)}. \quad (8.3)$$

Only update parameters that are free. Fixed initial probabilities, tied mixtures, forbidden transitions and zero-occupancy states require corresponding constraints or conventions. In particular, if α is constrained to be the stationary distribution of the *current* T , it cannot be optimized independently with these formulas. The package's default initial probabilities are fixed during an individual fit after their initialization. Dirichlet MAP parameters add $\alpha^{\text{prior}} - 1$ pseudo-counts on allowed entries where an interior MAP update exists.

8.4 Noise and likelihood terms

Under (4.1),

$$\mathcal{E}_{it}^m = \mathbb{E} \left(Y_{it}^m - \sum_k L_{ik}^m F_{tk}^m \right)^2 = \left(Y_{it}^m - \sum_k m_{ik}^{L,m} m_{tk}^{F,m} \right)^2 + \sum_k \left(M_{ik}^{L,m} M_{tk}^{F,m} - (m_{ik}^{L,m} m_{tk}^{F,m})^2 \right). \quad (8.4)$$

For a shared precision in an observed-entry group $\mathcal{G}$, its unpenalized maximizer is

$$\tau_{\mathcal{G}}^{\text{new}} = \frac{|\mathcal{G}|}{\sum_{(i,t) \in \mathcal{G}} \mathcal{E}_{it}}. \quad (8.5)$$

This presumes positive total residual and nonempty groups; unrestricted entry-specific variance estimation is generally not an appropriate model. With a Gamma prior of shape a_0 , rate b_0 , an independent variational factor is Gamma with shape $a_0 + |\mathcal{G}|/2$, rate $b_0 + \sum \mathcal{E}_{it}/2$. A Gibbs noise update uses realized squared residuals instead. Known precisions are not re-estimated.

The likelihood contribution to the ELBO is

$$\mathbb{E} \log p_m(Y^m | \cdot) = \frac{1}{2} \sum_{(i,t) \in \Omega_m} [\log \tau_{it}^m - \log(2\pi) - \tau_{it}^m \mathcal{E}_{it}^m] \quad (8.6)$$

for fixed precisions. Use $\mathbb{E} \log \tau$ and $\mathbb{E} \tau$, respectively, under an independent variational precision factor. For a loading mixture, $H(q_u) = -(1 - \gamma_u) \log(1 - \gamma_u) - \gamma_u \log \gamma_u + \gamma_u h(q_{u,1})$, on the disjoint spike/slab reference space. The tilted slab entropy can be evaluated from $\log q_{u,1}(v) = \log \varphi(v; m_0, v_0) + D_u(v) - \log I_0$. The HMM KL can be evaluated from its prior, normalized posterior and normal-means evidence; one need not force feature-position independence.

9 Simpler cases and a stronger variational alternative

9.1 Strict binary loadings: a closed-form scalar probability

Consider the different model $V_u \in \{0, 1\}$ with $g_u(1 | P_u) = \pi_u(P_u)$. Under independent Bernoulli variational factors, set $\gamma_u = q(V_u = 1)$. Then (4.8) gives

$$\boxed{\text{logit } \gamma_u = b_u - \frac{1}{2}a_u + \mathbb{E} \text{logit } \pi_u(P_u) + D_u(1) - D_u(0).} \quad (9.1)$$

The $-a_u/2$ term is essential because $1^2 - 0^2 = 1$. There is no continuous integral. Parent expectations may still require sums or approximation. For a small number of binary parents they can be enumerated exactly. The exact Gibbs probability uses fixed parents and children in the same expression. This is an excellent reference model for the notebook’s binary truth, but it is not the package’s continuous `gbinary` or CGB.

9.2 Linear-logistic CGB: a log-concave slab

Suppose each child gate has a linear logit in the candidate loading, $\pi_c(P_c[v]) = \text{sigmoid}(\beta_{cu}v + \zeta_c)$, where ζ_c depends on the other parents. Then under the same mean-field assumptions,

$$D'_u(v) = \sum_{c \in \mathcal{C}_u} \beta_{cu} \{\gamma_c - \mathbb{E} \pi_c(P_c[v])\}, \quad (9.2)$$

$$D''_u(v) = - \sum_{c \in \mathcal{C}_u} \beta_{cu}^2 \mathbb{E} \{\pi_c(P_c[v])(1 - \pi_c(P_c[v]))\} \leq 0. \quad (9.3)$$

Hence the CGB slab log posterior has second derivative $-1/v_0 + D''_u(v) < 0$. It is strictly log-concave and has one mode. One-dimensional quadrature or a suitable slice sampler is therefore simpler than for arbitrary neural logits. This result assumes parent-independent Gaussian slabs and linear logits; it is not a claim about the existing ReLU networks. Interactions can be introduced carefully, but an arbitrary nonlinear function of the target need not retain concavity.

For a child with positive β_{cu} , an inferred activation exceeding the gate’s predicted activation makes $D'_u(v) > 0$ for that contribution. It pushes the parent upward. This gives a direct algebraic interpretation of RNA-to-ATAC posterior information.

9.3 Keep all loadings of a cell together

A richer alternative replaces the product of scalar loading factors by $\prod_i q_i(A_i, R_i)$. It can retain both within-modality and cross-modality posterior dependence. With all other blocks fixed,

define

$$H_i^m = \sum_{t:(i,t) \in \Omega_m} \tau_{it}^m \mathbb{E}(F_{t,:}^m (F_{t,:}^m)^\top), \quad (9.4)$$

$$h_i^m = \sum_{t:(i,t) \in \Omega_m} \tau_{it}^m Y_{it}^m \mathbb{E} F_{t,:}^m. \quad (9.5)$$

Here $F_{t,:}^m$ is understood as a column vector. The unrestricted row coordinate is

$$q_i^*(a, r) \propto p(a, r \mid \theta) \exp \left\{ -\frac{1}{2} a^\top H_i^A a + (h_i^A)^\top a - \frac{1}{2} r^\top H_i^R r + (h_i^R)^\top r \right\}. \quad (9.6)$$

For strict binary loadings this can be normalized by enumerating $2^{K_A+K_R}$ states per cell. With five factors in each modality this is 1,024 states per cell; feasibility still depends on network evaluation and the number of cells and iterations. Continuous CGB loadings make the row integral multivariate and substantially harder.

Row dependence changes the factor pseudo-data. In particular, replace the factor numerator by

$$b_{tk}^{F,m} = \sum_{i:(i,t) \in \Omega_m} \tau_{it}^m \left[Y_{it}^m \mathbb{E} L_{ik}^m - \sum_{h \neq k} \mathbb{E}(L_{ik}^m L_{ih}^m) \mathbb{E} F_{th}^m \right]. \quad (9.7)$$

The denominator is $\sum_i \tau_{it}^m \mathbb{E}[(L_{ik}^m)^2]$, using the second moment rather than $(\mathbb{E} L_{ik}^m)^2$. The HMM form remains intact, but the existing mean-field residual formula is insufficient: within-row cross moments are needed. Likewise the residual second moment must be evaluated with those cross moments. Conditional on fixed F , exact binary row Gibbs uses the same construction with realized factor products in place of their expectations.

10 Qualifications needed for a coherent implementation

10.1 Covariate standardization

The package's `standard_scale` computes a column mean and standard deviation over the current input rows. If these statistics are recomputed as functions of latent loading columns *inside the probability model*, changing one A_{ik} changes transformed covariates for other cells. Later conditional priors in those cells then enter its Markov blanket too. The cell-local equations above would be incomplete.

For a first joint specification, choose fixed offsets/scales, for example from an initialization, and regard them as known transformations. Alternatively define a model with explicit transformation parameters and optimize them under one objective. Recomputing transformations between steps without accounting for their effect does not preserve a fixed joint inference target.

10.2 The current sparsity penalty is not automatically a parameter prior

With fixed covariates the CGB loss adds $(\lambda - 1) \sum_i \log(1 - \pi_{iu}(X_i))$ to the log-likelihood criterion. If the covariates become latent parents, that expression depends on latent variables, not only on neural parameters. One possible explicitly regularized variational criterion is

$$\mathcal{L}_{\text{reg}}(q, \Theta) = \mathcal{L}(q, \Theta) + (\lambda - 1) \sum_u \mathbb{E}_q \log(1 - \pi_u(P_u)). \quad (10.1)$$

Its loading-coordinate update must include the added terms whenever the target is a parent. It is not the ELBO of the unpenalized joint model. To interpret the same factor as a new

normalized tilted joint density, its normalizing constant must be included; this generally depends on parameters and affects their learning. Simply multiplying each child’s conditional by a function of its parents and renormalizing that conditional cancels the function, so it does not give the intended regularization.

A clean initial derivation therefore uses $\lambda = 1$, with optional proper priors or explicit regularizers on *parameters alone*. HMM Dirichlet parameter priors have a different, direct expected-count interpretation. The same numerical penalty value across different solvers need not mean the same statistical regularization.

10.3 Ordering, rescaling, support changes and pruning

For each modality, $L_k \mapsto cL_k, F_k \mapsto F_k/c$ preserves the fitted matrix for $c \neq 0$. It need not preserve the complete prior or the joint objective. Prior parameters, neural input transformations and HMM supports must be transformed consistently before treating rescaling as a mere reparameterization. Fix a scale convention for interpretation and initialization.

Order matters in (2.4). Pruning or permuting a component changes the identities and dimensions of later predictors. Such changes require model reconstruction and objective evaluation, not only deletion of one loading column. A deterministic neural conditional can also create nearly hard dependencies and poor single-coordinate mixing; bounded gate probabilities and row-block inference can help, but change the approximation or admissible parameter space and should be specified.

Moving the location of a point mass while holding its old posterior fixed can make the expected prior log density minus infinity. One should not apply an ordinary fixed-posterior M-step blindly to moving atoms. Keeping HMM atoms and grids fixed is the simplest derivation. Jointly refitting a factor posterior and its HMM under a changed support can instead be considered as an EB block proposal, assessed using its collapsed marginal likelihood and the consistent joint objective. The distinction is between a valid evaluated block proposal and an assumed fixed-support EM update.

10.4 A mathematically defined iteration

For the unpenalized fixed-rank, fixed-scaling model, one possible schedule is:

1. Initialize all loading distributions, factor HMM posteriors and parameters.
2. Update each ATAC loading using (4.8), including later ATAC and RNA children; retain its spike and slab representation.
3. Update each RNA loading, including its later RNA children.
4. Update each factor sequence by (7.4) and forward/backward.
5. Update shared loading-prior, HMM and noise parameters using the common objective. A generalized EM step must increase that objective, even if not maximizing it completely.
6. Evaluate the *single* objective (4.3) at the current state. Check convergence and numerical accuracy; repeat if needed.

Other scan orders are possible. Exact coordinate steps optimize the same objective; local optima and rates of convergence can still depend on order and initialization. Keeping old per-factor KL values after changing their parents does not evaluate the current joint objective. A Gibbs version samples full conditionals instead and targets a posterior rather than an ELBO.

11 How useful this could be

11.1 What the construction could provide

1. **Bidirectional use of paired data.** RNA can resolve ambiguous ATAC loadings and ATAC can resolve ambiguous RNA loadings under one directed prior. A separate reverse prior is not required.
2. **Different ranks and modality-specific programs.** This model does not force $A = R$ or equal ranks. It models relationships among activity patterns while preserving modality-specific feature factors.
3. **A precise extension of EBNM modularity.** The ordinary Gaussian normal-means reduction remains, and the HMM factor block remains reusable. Equation (6.10) identifies exactly what a new loading solver must add.
4. **A testable objective.** The derivation separates a joint-model inference algorithm from a useful but uncalibrated covariate-exchange heuristic. It specifies what must be monitored and which terms a proposed shortcut drops.

This makes the derivation useful before investing in a general neural solver. It does not establish novelty, identifiability, causal direction or guaranteed improvement. EBMF’s normal-means reduction is established work [3]; covariate-modulated priors are the cEBMF setting [4]. The present note derives their ordered joint extension for this project.

11.2 Why the supplied simulation is informative

The noiseless RNA profiles distinguish the four binary ATAC states. For the three RNA blocks at positions 200–249, 500–549 and 700–749 (Python indexing), the mapping is

ATAC loading 1	ATAC loading 2	RNA block 1	RNA block 2	RNA block 3
0	0	1	1	0
0	1	1	1	1
1	0	0	1	1
1	1	1	0	1

Thus RNA has information about the ATAC state that could improve ATAC inference when its own observations are noisy. The two independent noise realizations make combining their evidence sensible under the correctly specified joint model. The four one-hot RNA loading columns used to generate the data do not force fitted RNA rank four: the noiseless matrix has rank three, and a three-block factorization is another representation.

Finite-sample estimation, misspecified gates, incorrect pairing and local optima can make sharing harmful. Under a correct known model, conditioning on more data cannot increase the expected Bayes squared-error risk of a posterior mean; that statement does not guarantee improvement in every realized dataset or for fitted empirical Bayes/variational approximations. More concretely, for a square-integrable target U , the conditional-expectation identity gives

$$\mathbb{E}\{(U - \mathbb{E}[U \mid Y_A])^2\} - \mathbb{E}\{(U - \mathbb{E}[U \mid Y_A, Y_R])^2\} = \mathbb{E}\{(\mathbb{E}[U \mid Y_A, Y_R] - \mathbb{E}[U \mid Y_A])^2\} \geq 0. \quad (11.1)$$

11.3 The most informative first comparison

A small strict-binary version with exact enumeration can validate the posterior-feedback equations without neural or quadrature uncertainty. A linear-logistic CGB version can then isolate

the continuous-loading problem while retaining a log-concave scalar slab. A flexible neural version becomes more informative after those reference cases are understood.

Useful comparisons include independent fits, each fixed one-direction fit, the covariate-exchange heuristic, and the joint-target method. Evaluate both modalities, using held-out observations excluded from *every* preprocessing and fitting stage that could leak those values. In simulations also compare against the noiseless matrices and assess loading uncertainty where a common scale/order is defined. Check shuffled cell pairing, multiple initializations, different scan orders, and weak or absent cross-modality dependence. Section 12 places these comparisons in the existing literature; the derivation alone does not establish methodological novelty.

12 Comparison with other multi-omics methods

12.1 Scope and notation

This section compares the proposed model with representative multi-omics methods, using the literature reviewed on 14 September 2026. It is a comparison of statistical structure and inferential requirements, not an empirical performance ranking or an exhaustive novelty review. Throughout, L^m denotes sample loadings and F^m denotes feature factors; some cited methods use the opposite terminology.

Three questions must be separated: which finite signal matrices can be represented, which joint distributions of sample programs can be modeled, and how accurately those distributions can be learned and their posteriors approximated. A more flexible conditional family need not improve finite-sample prediction, calibration, or computation.

For reference, a shared-score linear predictor has the form

$$Y^m = ZW_m^\top + E^m, \quad (12.1)$$

whereas a shared/private decomposition can be written schematically as

$$Y^m = SU_m^\top + V_mB_m^\top + E^m. \quad (12.2)$$

Here Z, S are sample-score matrices shared across views and V_m contains view-specific sample scores. The proposed formulation instead uses $Y^A = AF_A^\top + E^A$, $Y^R = RF_R^\top + E^R$, and the normalized conditional prior (2.4). These schematic matrix equations do not identify methods' complete model classes: their priors, likelihoods, rank constraints, and identifiability restrictions can differ.

12.2 Structural comparison

Table 2 summarizes the cited formulations. A mechanism absent from a standard formulation may be possible in an extension. In particular, neither modality-specific programs nor joint borrowing of information is unique to the proposed hierarchy.

Table 2: Representative multi-omics methods and their relation to the proposed conditional loading model. Dates refer to the cited papers.

Method	Native model structure	Relation to this derivation
JIVE (2013); AJIVE (2018)	Decompose joint and individual low-rank sample subspaces. AJIVE uses subspace angles to estimate shared variation.	Original formulations describe linear shared structure rather than learned nonlinear conditional distributions between distinct sample programs [6, 7].
MOFA+ (2020)	Shared sample scores and view-specific feature weights, with shrinkage, groups, missing data, and several observation likelihoods.	Already supports shared, view-specific and subset-active factors. The standard score-prior construction differs from the autoregressive conditional loading prior (2.4) [5].
ProJIVE (2026)	Probabilistic joint and individual factorization, fitted by maximum likelihood with EM.	A probabilistic extension of JIVE already exists; probability modeling alone is not a distinguishing contribution [12].
SLIDE (2019); BASS (2016)	Components shared across subsets of views. SLIDE uses structured sparsity; BASS uses Bayesian structured shrinkage.	Partial sharing and sparse/dense program structure are established capabilities, distinct from conditional dependence among programs [8, 9].
BIDIFAC+ (2022); EV-BIDIFAC (2024)	Low-rank modules shared across subsets of linked row and column groups; EV-BIDIFAC uses empirical variational Bayes shrinkage.	Close empirical Bayes integration baselines, including missing-data imputation. The proposed additional mechanism is learning conditional program distributions [10, 11].
SIFA (2017)	Joint and individual factors partially informed by observed auxiliary covariates through nonparametric functions.	Nonlinear covariate-informed factors already exist. Uncertain latent parents and the required posterior feedback distinguish the present inference problem [15].
MuVI (2023); SOFA (2026)	MuVI incorporates feature-set information through structured shrinkage. SOFA links guided factors to sample-level guide variables and retains unguided factors.	Relevant comparisons when biological feature sets or sample annotations are supplied. They address other uses of prior information [16, 17].
MEFISTO (2022)	Gaussian-process structure on sample factors along continuous covariates such as time and space.	Sample-axis covariance differs from the feature-axis HMM in Section 7. Structured factor priors themselves are not new [18].
CAVACHON (2024 preprint)	A supplied modality DAG organizes a hierarchical VAE with mixture innovation latents and ancestor-aware decoders.	Close prior art for biological hierarchy and conditional independence; its decoder construction differs from normalized scalar loading priors with linear feature reconstruction [13].
HALO (2025)	RNA/ATAC representations linked through learned mappings and causal constraint losses; an interpretable nonlinear additive decoder.	Hierarchical RNA–ATAC analysis and interpretable decoding already exist. Its objective differs from the normalized joint-prior construction here [14].
MultiVI (2023); GLUE (2022)	MultiVI integrates paired and unpaired RNA/ATAC data with a multimodal VAE. GLUE links modality embeddings through a feature guidance graph.	Important single-cell comparators with different information requirements and observation models. The present paired Gaussian model has narrower scope [19, 20].

12.3 Observed side information and uncertain latent parents

The model can combine fixed information x_i^A, x_i^R with latent programs:

$$\begin{aligned} A_{ik} &| A_{i,<k}, x_i^A \sim g_k^A(\cdot | A_{i,<k}, x_i^A), \\ R_{ij} &| R_{i,<j}, A_i, x_i^R \sim g_j^R(\cdot | R_{i,<j}, A_i, x_i^R). \end{aligned} \quad (12.3)$$

This makes explicit the fixed covariates suppressed in most earlier equations. In (5.2), replace every $g_u(v | P_u)$ by $g_u(v | P_u, x_u)$, and every child density by $g_c(V_c | P_c[v], x_c)$. Average over uncertain latent inputs, while holding the observed x 's fixed. The fixed inputs can modulate child feedback and do not disappear from those density evaluations.

EBMF's reduction to normal-means problems is established [3], and flexible priors informed by fixed side information are already part of cEBMF [4]. Thus the added inferential problem is not simply concatenating more covariates. It is that changing a latent loading changes other factors' priors, requiring both uncertain-parent expectations and child contributions under one objective. The same framework can condition on observed context and latent programs simultaneously; it need not choose between them.

12.4 Comparison with hierarchical neural models

CAVACHON's generative equations use Gaussian-mixture innovation variables and feed ancestor representations into the observation decoder [13]. In the present construction, the hierarchy acts on scalar loading distributions, while $F^m(L_i^m)^\top$ retains a linear additive feature representation. These architectures impose different assumptions; neither comparison alone proves greater distributional capacity.

HALO's displayed generative specification uses Gaussian latent priors, with additional mapping and causal-constraint terms in its training objective (Methods, Equations 17–24) [14]. It should therefore not be identified with exactly the normalized conditional-prior model in (2.4). Conversely, hierarchical RNA–ATAC modeling and interpretability alone cannot distinguish this proposal from HALO.

The potential contribution is a particular statistical model and inference construction: normalized conditional program distributions, explicit additive feature factors, and EBNM-based updates across multiple prior families. Accurate uncertainty, computational advantages and biological usefulness remain empirical questions.

12.5 Reconstruction capacity versus conditional dependence

Shared-score representation of any fitted pair. Let $M^A = AF_A^\top$, $M^R = RF_R^\top$, and set

$$Z = [A \ R], \quad W_A = [F_A \ 0], \quad W_R = [0 \ F_R]. \quad (12.4)$$

Then $M^A = ZW_A^\top$ and $M^R = ZW_R^\top$. A shared-score representation with at most $K_A + K_R$ columns can therefore reproduce these finite fitted matrices exactly. This is a statement about matrix parameterization, not equality of their latent prior distributions or of the full model classes. It rules out a universal reconstruction advantage based solely on separate loading matrices.

Linear Gaussian hierarchy as shared/private structure. For column-vector loadings, suppose $a \sim \mathcal{N}(0, \Sigma_A)$ and $r = Ba + \eta$, with $\eta \sim \mathcal{N}(0, \Sigma_\eta)$ independent of a . The observation equations become

$$y^A = F_A a + \epsilon_A, \quad y^R = F_R B a + F_R \eta + \epsilon_R. \quad (12.5)$$

Thus a supplies shared variation and η supplies RNA-specific variation. For positive-definite covariance matrices, whitening gives independent standard Gaussian scores with transformed feature weights. At the observation-distribution level this is shared/private linear Gaussian structure, although method-specific sparsity or orthogonality constraints can still differ.

Nonlinear dependence. Let A_1, A_2 be independent Bernoulli(1/2) variables and $R = A_1 \text{ xor } A_2$. Then $\mathbb{E}R = 1/2$ and $\text{Cov}(A_1, R) = \text{Cov}(A_2, R) = 0$. The optimal affine predictor from these two inputs is $1/2$, with population squared-error risk $1/4$; the nonlinear oracle

$$\mathbb{E}(R \mid A_1, A_2) = A_1 + A_2 - 2A_1A_2 \quad (12.6)$$

has zero risk. A neural CGB gate can approximate this relationship as its activation probabilities approach zero/one and its active slab concentrates near one. It is not exactly a binary prior at finite positive slab variance.

This example establishes a limitation of affine prediction using the specified two inputs. An expanded basis containing A_1A_2 represents the mean with a linear readout, and non-Gaussian latent models can have nonlinear posterior predictors. These calculations are not fitted MOFA+ or JIVE error rates and do not prove that every configuration of those methods fails on XOR.

Conditional variance can be useful without a mean association. Suppose $R \mid A = a \sim \mathcal{N}(0, v(a))$ and $T = R + \varepsilon$ with independent $\varepsilon \sim \mathcal{N}(0, s^2)$. Then $\mathbb{E}(R \mid A) = 0$, but

$$\mathbb{E}(R \mid T, A = a) = \frac{v(a)}{v(a) + s^2}T, \quad \text{Var}(R \mid T, A = a) = \frac{v(a)s^2}{v(a) + s^2}. \quad (12.7)$$

ATAC can inform the amount of RNA shrinkage even without predicting its conditional mean from ATAC alone. This is relevant to the centered conditional scale-mixture families in Section 5.4. Predictive dispersion and calibration should therefore supplement cross-modal mean prediction in a benchmark.

12.6 Implications for inference and empirical comparison

Full joint inference allows evidence from either modality to update the other, including when the generative prior is directed. This is not unique to the present model: shared latent-variable models also borrow information from all contributing views. The relevant question is whether the proposed conditional distributions provide better assumptions for a particular scientific problem.

A dense graph with unrestricted conditionals can express any joint law by the chain rule in different orders. It does not identify biological causality. Sparse parents, restricted conditional families and anchored factor definitions can make a hierarchy more informative, but their assumptions and sensitivity require assessment. Scale, permutation, sign and possible rotation ambiguities must be accounted for when interpreting dependencies between estimated programs.

An informative comparison should include independent cEBMF, one-way plug-in conditioning in each direction, within-modality hierarchy only, a linear conditional model, and the joint-target updates derived here. A two-stage nonlinear map between independently fitted loadings helps separate the benefit of flexible regression from that of joint inference. External comparators should cover both linear/empirical Bayes methods and hierarchical neural models in Table 2.

Evaluate held-out prediction and calibration separately for both modalities, with ranks and tuning budgets treated fairly. Vary noise asymmetry, nonlinear gating, variance-only dependence, incorrect or absent dependence, and shuffled pairing. Hold feature priors fixed in loading-model

ablations so that HMM smoothing is not mistaken for a benefit of the hierarchy. The HMM uses feature adjacency, whereas fixed genomic distances or sample coordinates are different inputs.

For new-sample prediction, exclude held-out observations from preprocessing, factor fitting, covariate construction and conditional-network training. Separate entry-wise completion from entirely missing-modality prediction, and split by donors or studies when that is the unit of generalization. Compare predictions for the same observed quantity and units; raw ELBOs across different models or Gaussian transformed-data losses against count likelihoods are not direct performance comparisons. Original deterministic decompositions do not provide a predictive density without an additional model.

The present Gaussian observation model does not automatically inherit count-data, batch-correction or unpaired-integration capabilities from the cited methods. Likewise, the general coordinate identity and Metropolis rule are established tools [2, 1]; their application is not itself a novel general theorem. A methodological contribution would require a coherent implemented solver, evidence that its approximation is accurate, and a demonstrated advantage of the conditional-program assumptions in appropriate settings.

A Deriving the likelihood coefficients

For one observed entry, write the contribution excluding the target as $S_{-k} = \sum_{h \neq k} L_{ih} F_{th}$. Expanding gives

$$(Y - vF_{tk} - S_{-k})^2 = v^2 F_{tk}^2 - 2vF_{tk}(Y - S_{-k}) + (Y - S_{-k})^2. \quad (\text{A.1})$$

Under (4.1), $\mathbb{E}F_{tk}^2 = M_{tk}^F$ and $\mathbb{E}[F_{tk}(Y - S_{-k})] = m_{tk}^F(Y - \sum_{h \neq k} m_{ih}^L m_{th}^F)$. Multiplication by $-\tau/2$ and summation over observed features gives (4.6)–(4.7). For a factor update, interchange loading and factor roles. If loading columns are dependent within a row, $\mathbb{E}(L_{ik} L_{ih})$ cannot be factorized, yielding (9.7).

B Two-node binary illustration of posterior feedback

Let $A, R \in \{0, 1\}$, $A \sim \text{Bernoulli}(p)$, $R | A = a \sim \text{Bernoulli}(\rho_a)$. Suppose the ATAC likelihood alone contributes log odds λ_A for $A = 1$ against $A = 0$. Conditional on $R = r$,

$$\text{logit } p(A = 1 | R = r, Y_A) = \text{logit } p + \lambda_A + r \log \frac{\rho_1}{\rho_0} + (1 - r) \log \frac{1 - \rho_1}{1 - \rho_0}. \quad (\text{B.1})$$

For mean-field $q(R = 1) = \gamma_R$, replace r by γ_R to get the CAVI update. This is an expected log message. By contrast, the exact marginal posterior for A given RNA data with likelihoods $L_R(r) = p(Y_R | R = r)$ has the RNA likelihood ratio

$$\frac{(1 - \rho_1)L_R(0) + \rho_1 L_R(1)}{(1 - \rho_0)L_R(0) + \rho_0 L_R(1)}. \quad (\text{B.2})$$

The log of this ratio is generally not the mean-field expected-log term. This example distinguishes an exact scalar full conditional, an exact *marginal* posterior, and a variational coordinate update. They are three different quantities even though they arise from the same joint model.

C Verification and package correspondence

The formulas were checked algebraically by expansion and by the limiting cases listed below. A companion numerical check of this note, rather than an implementation in the package, compares small finite-state and one-dimensional calculations:

1. No children and fixed parents recover ordinary CGB posterior moments.
2. A one-child tilted CGB conditional agrees with direct numerical integration of its spike and slab joint likelihood.
3. A binary two-node full conditional agrees with enumeration of all four joint states. Its mean-field update agrees with direct coordinate maximization and increases the same finite-state ELBO.
4. The EBNM-proposal Metropolis kernels satisfy detailed balance, and a complete two-coordinate sweep preserves the enumerated binary joint posterior.
5. The augmented EMDN update agrees with direct integration of the expected joint log density when uncertain parents change weights, means and variances, and an EMDN child contributes a downstream term.
6. The positive-slab evidence in (6.11) agrees with direct convolution, and the example’s RNA block mapping agrees with its code.

These checks validate the local identities, not convergence or performance of an unimplemented full joint solver. The associated numerical results are reported in the companion file `atac_rna_derivation_checks.txt`. The companion script `verify_atac_rna_derivation.py` reproduces these checks using Python, NumPy and SciPy. It is a verification script for this document, not a factorization solver.

The inspected source locations are relative to the repository root:

- `src/cebmf_torch/cebnm/cov_gb_prior.py`
Gate-only covariate dependence, signed global Gaussian slab, local fitting criterion and penalty.
- `src/cebmf_torch/cebmf/cebmf.py`
Loading and factor pseudo-data, posterior moments, covariate construction, local KL accounting and noise updates.
- `src/cebmf_torch/cebnm/hmm.py`
Unit-spaced HMM normal-means model, state mixtures, signed/one-sided variants and smoothing.
- `src/cebmf_torch/utils/standard_scaler.py`
Standardization recomputed from the current covariate matrix.
- `src/cebmf_torch/priors/learned.py`
Learned registry names and the common moment-returning adapter. The CASH, LCASH, sharp CGB and EMDN solver modules define the conditional families mapped in Section 5.4.
- `docs/source/ordered_loading_priors.rst` and `docs/source/hmm_priors.rst`
Current user-facing model semantics.

The local CGB solver’s moment-based slab-variance estimator is not being identified with the joint latent-variable M-step (8.2). This note defines the latter independently. No package implementation is changed by this derivation.

References

- [1] W. K. Hastings (1970). *Monte Carlo sampling methods using Markov chains and their applications*. *Biometrika* 57(1), 97–109. [doi:10.1093/biomet/57.1.97](https://doi.org/10.1093/biomet/57.1.97). Reference for the Metropolis–Hastings acceptance rule used in Section 5.3; the EBNM cancellation is derived here.
- [2] D. M. Blei, A. Kucukelbir and J. D. McAuliffe (2017). *Variational Inference: A Review for Statisticians*. *Journal of the American Statistical Association* 112(518), 859–877. [arXiv:1601.00670](https://arxiv.org/abs/1601.00670). Reference for the generic ELBO and mean-field coordinate identity.
- [3] W. Wang and M. Stephens (2021). *Empirical Bayes Matrix Factorization*. *Journal of Machine Learning Research* 22(120), 1–40. <https://www.jmlr.org/papers/v22/20-589.html>. Reference for EBMF and its normal-means reduction.
- [4] W. R. P. Denault, K. Tayeb, P. Carbonetto, J. Willerscheid and M. Stephens (2025). *Covariate-moderated Empirical Bayes Matrix Factorization*. [arXiv:2505.11639v2](https://arxiv.org/abs/2505.11639v2). Reference for the covariate-moderated framework discussed in this project. The joint-loading equations in this note are derived for the proposed hierarchy; they are not attributed to this reference.
- [5] Argelaguet et al. (2020). MOFA+: a statistical framework for comprehensive integration of multi-modal single-cell data. *Genome Biology* 21, 111. <https://doi.org/10.1186/s13059-020-02015-1>
- [6] Lock, Hoadley, Marron and Nobel (2013). Joint and Individual Variation Explained (JIVE) for integrated analysis of multiple data types. *Annals of Applied Statistics* 7, 523–542. <https://pmc.ncbi.nlm.nih.gov/articles/PMC3671601/>
- [7] Feng, Jiang, Hannig and Marron (2018). Angle-based joint and individual variation explained. <https://arxiv.org/abs/1704.02060>
- [8] Gaynanova and Li (2019). Structural learning and integrative decomposition of multi-view data. *Biometrics* 75, 1121–1132. <https://arxiv.org/abs/1707.06573>
- [9] Zhao, Gao, Mukherjee and Engelhardt (2016). Bayesian group factor analysis with structured sparsity. *Journal of Machine Learning Research* 17. <https://www.jmlr.org/papers/volume17/14-472/14-472.pdf>
- [10] Lock et al. (2022). Decomposition and imputation of bidimensionally linked matrices. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9060567/>
- [11] Lock (2024). Empirical Bayes Linked Matrix Decomposition. *Machine Learning* 113, 7451–7477. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11698509/>
- [12] Murden et al. (2026). Probabilistic Joint and Individual Variation Explained (ProJIVE) for Data Integration. *Journal of Computational and Graphical Statistics*; arXiv version accessed. <https://arxiv.org/abs/2603.12351>
- [13] Hsieh et al. (2024). CAVACHON: a hierarchical variational autoencoder to integrate multi-modal single-cell data. Preprint, arXiv:2405.18655v1. Earlier related paper presented at NeurIPS LMRL Workshop 2022. <https://arxiv.org/html/2405.18655v1>
- [14] Mao et al. (2025). HALO: hierarchical causal modeling for single cell multi-omics data. *Nature Communications* 16, 8892. <https://www.nature.com/articles/s41467-025-63921-1>

- [15] Li and Jung (2017). Incorporating Covariates into Integrated Factor Analysis of Multi-View Data. *Biometrics* 73, 1433–1442. <https://arxiv.org/abs/1703.05794>
- [16] Qoku and Buettner (2023). Encoding Domain Knowledge in Multi-view Latent Variable Models: A Bayesian Approach with Structured Sparsity. *AISTATS*, PMLR 206. <https://proceedings.mlr.press/v206/qoku23a.html>
- [17] Capraz et al. (2026). Semi-supervised Omics Factor Analysis (SOFA) disentangles known and latent sources of variation in multi-omic data. *Nature Communications* 17, 8725. Published 20 August 2026. <https://www.nature.com/articles/s41467-026-74694-6>
- [18] Velten et al. (2022). Identifying temporal and spatial patterns of variation from multimodal data using MEFISTO. *Nature Methods* 19, 179–186. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8828471/>
- [19] Ashuach et al. (2023). MultiVI: deep generative model for the integration of multimodal data. *Nature Methods*. <https://www.nature.com/articles/s41592-023-01909-9>
- [20] Cao and Gao (2022). Multi-omics single-cell data integration and regulatory inference with graph-linked embedding. *Nature Biotechnology*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9546775/>